

National University of Computer and Emerging Sciences, Lahore Campus

	Course Name:	Data Analysis and Visualization	Course Code:	DS3001
	Degree Program:	BS (Data Science)	Semester:	Fall 2023
	Exam Duration:	60 Minutes	Total Marks:	50
	Paper Date:	2 nd -October-2023	Weight	15%
	Section:	ALL	Page(s):	5 + 1 (Rough Page)
	Exam Type:	Mid-1		

Name: _____ Roll No. _____ Section: _____

- Instruction/Notes:**
- Attempt all questions on the provided question paper.
 - Show all calculations, direct answers are not acceptable
 - Space for rough work is provided at the end of the paper.
 - Even if you do use extra rough sheets, they should **NOT** be attached with final paper.

Question #	1	2	3	Total
Total Marks	5	20	25	50
Obtained Marks				
CLO #				

Question 1:

You are provided with a Python web scraping code snippet and the corresponding HTML snippet. Your task is to dry run the code and write down the expected output.

```
import requests
from bs4 import BeautifulSoup

url = "http://example.com/data"
response = requests.get(url)
html_content = response.content

soup = BeautifulSoup(html_content, 'html.parser')
table = soup.find('table', class_='data-table')

data = []
for row in table.find_all('tr'):
    cols = row.find_all(['th', 'td'])
    cols = [col.text.strip() for col in cols]
    data.append(cols)

for row in data:
    print(row)
```

```
<!DOCTYPE html>
<html>
  <head>
    <title>Example Data</title>
  </head>
  <body>
    <table class="data-table">
      <tr>
        <th>Country</th>
        <th>Population</th>
      </tr>
      <tr>
        <td>USA</td>
        <td>328 million</td>
      </tr>
      <tr>
        <td>China</td>
        <td>1.4 billion</td>
      </tr>
      <tr>
        <td>India</td>
        <td>1.3 billion</td>
      </tr>
    </table>
  </body>
</html>
```

Output:

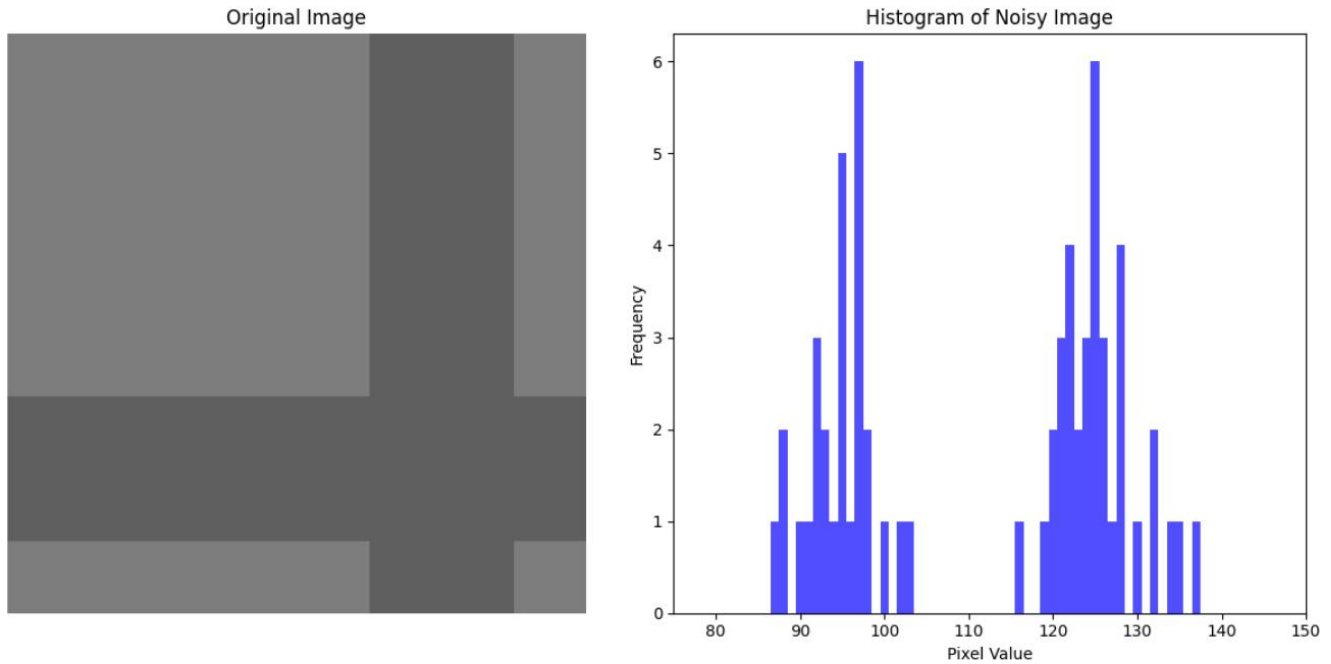
Question # 2

You have been provided with a dataset containing information about monthly sales figures for a small business. However, the dataset has some issues that need to be addressed before analysis. Your task is to identify the noisy values using IQR method and smooth the noise in sales using regression model.

Month	Sales	Cleaned Sales
1	1300	
2	1500	
3	1520	
4	1540	
5	1560	
6	1600	
7	1610	
8	1625	
9	1650	
10	3200	
11	2500	

Question # 3

a) Consider the following original image and histogram of noisy image



i. Identify the type of noise added in the original image?

ii. What type of filter you will choose to mitigate this noise? Be specific.

b) Consider a 4x4 gray image and a 3x3 filter:

$$\text{Gray image} = \begin{bmatrix} 80 & 10 & 20 & 90 \\ 170 & 8 & 240 & 15 \\ 20 & 150 & 163 & 40 \\ 12 & 34 & 40 & 70 \end{bmatrix},$$

$$\text{filter} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ -1 & -1 & -1 \end{bmatrix}$$

i. Convolve the gray image with the given filter with replicate padding, and show your result (whose size should be 4x4)

- ii. Discuss the applications of the given filter based on your knowledge.

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Rough Work